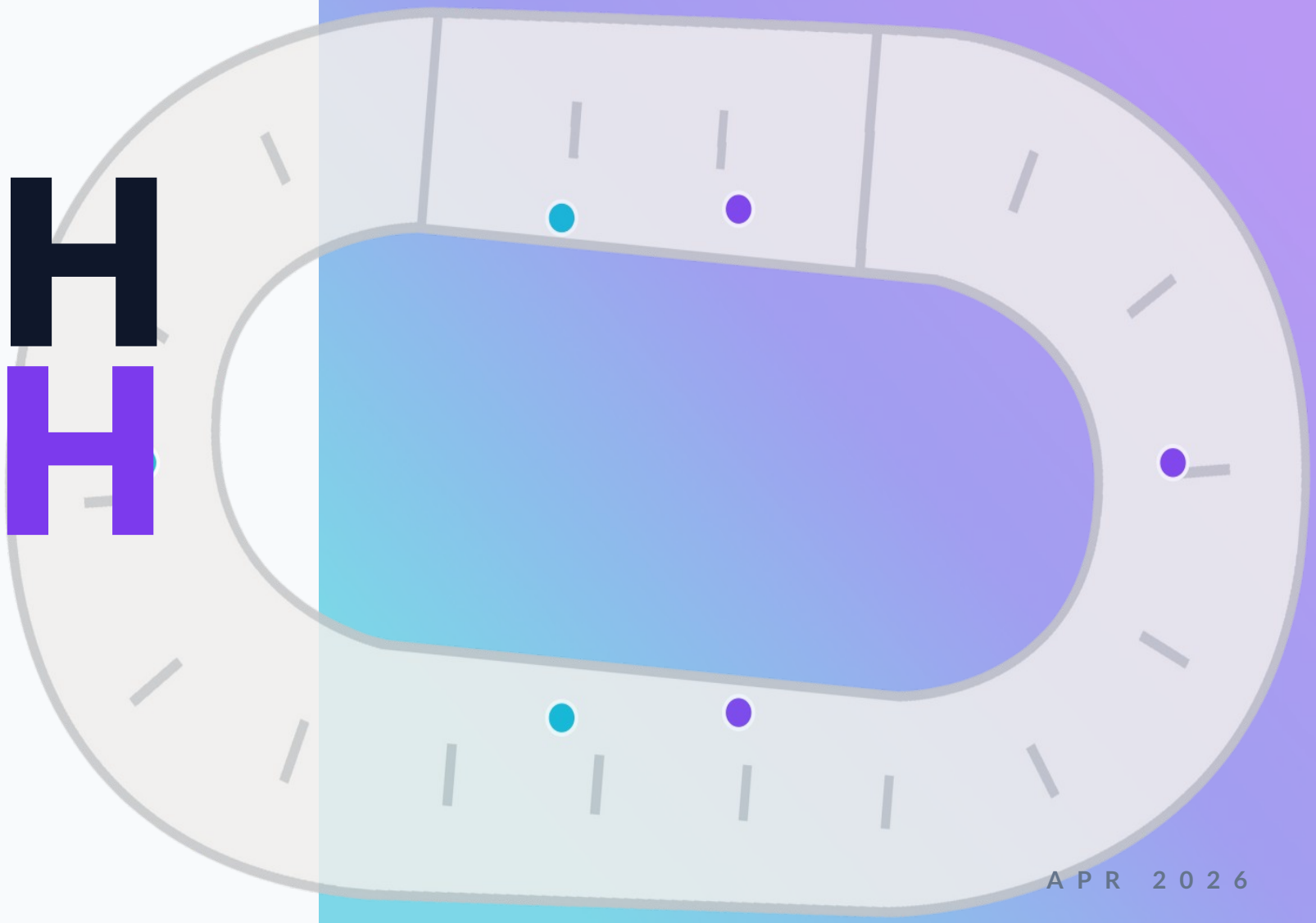
BENCH WITCH



DESIGN DECK · v2

APR 2026

Coaches can see it. They can't track it.

WHERE

Bench coaches notice patterns — "their jammer always blows out the inside on turn 4" — but have no way to capture that data live.

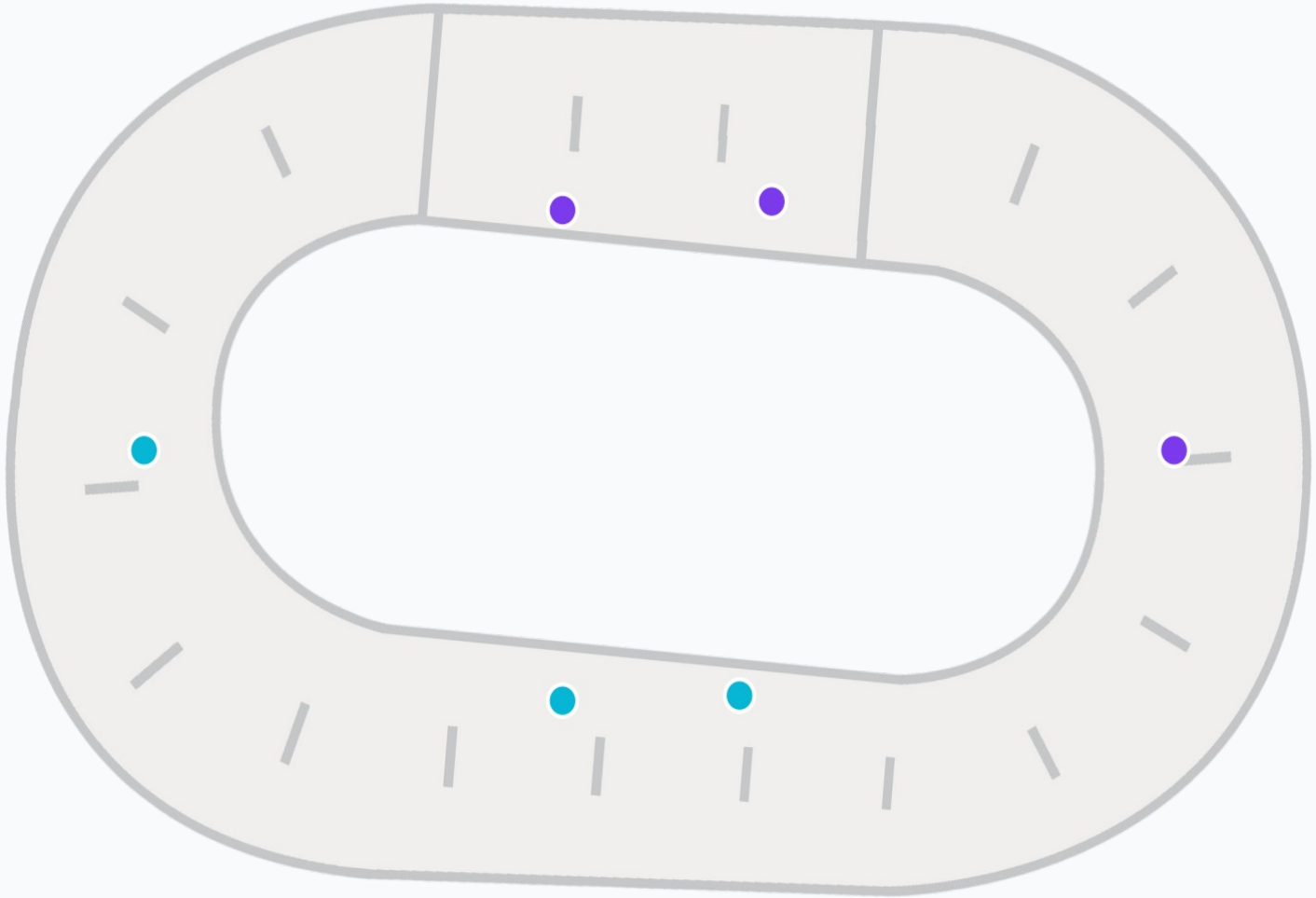
WHEN

Two-minute jams. No time to scribble notes, draw diagrams, or wrangle a stat sheet. Pen-and-paper kills you.

PROOF

Without spatial data, scouting is gut feel. Heat maps turn coach instinct into evidence skaters can train against.

One tap on the track. That's the whole UI.



1 **TAP**

Mark exactly where each jammer pops the pack.

2 **CLASSIFY**

Lead. Star pass. Points. Three buttons, one jam.

3 **SEE**

Heat maps reveal the patterns instantly.

4 **SAVE**

Every jam, every game, in your D1 database.

What a coach does, every two minutes.

JAM STARTS

1

App auto-creates the next jam.

TAP L

2

Whoever gets lead — one button per team.

TAP TRACK

3

Drop a dot where each jammer exited the pack.

TAP P (if)

4

Star pass — auto-shifts that jammer to Pivot slot.

TAP POINTS

5

Cumulative points per jam — one tap per pass.

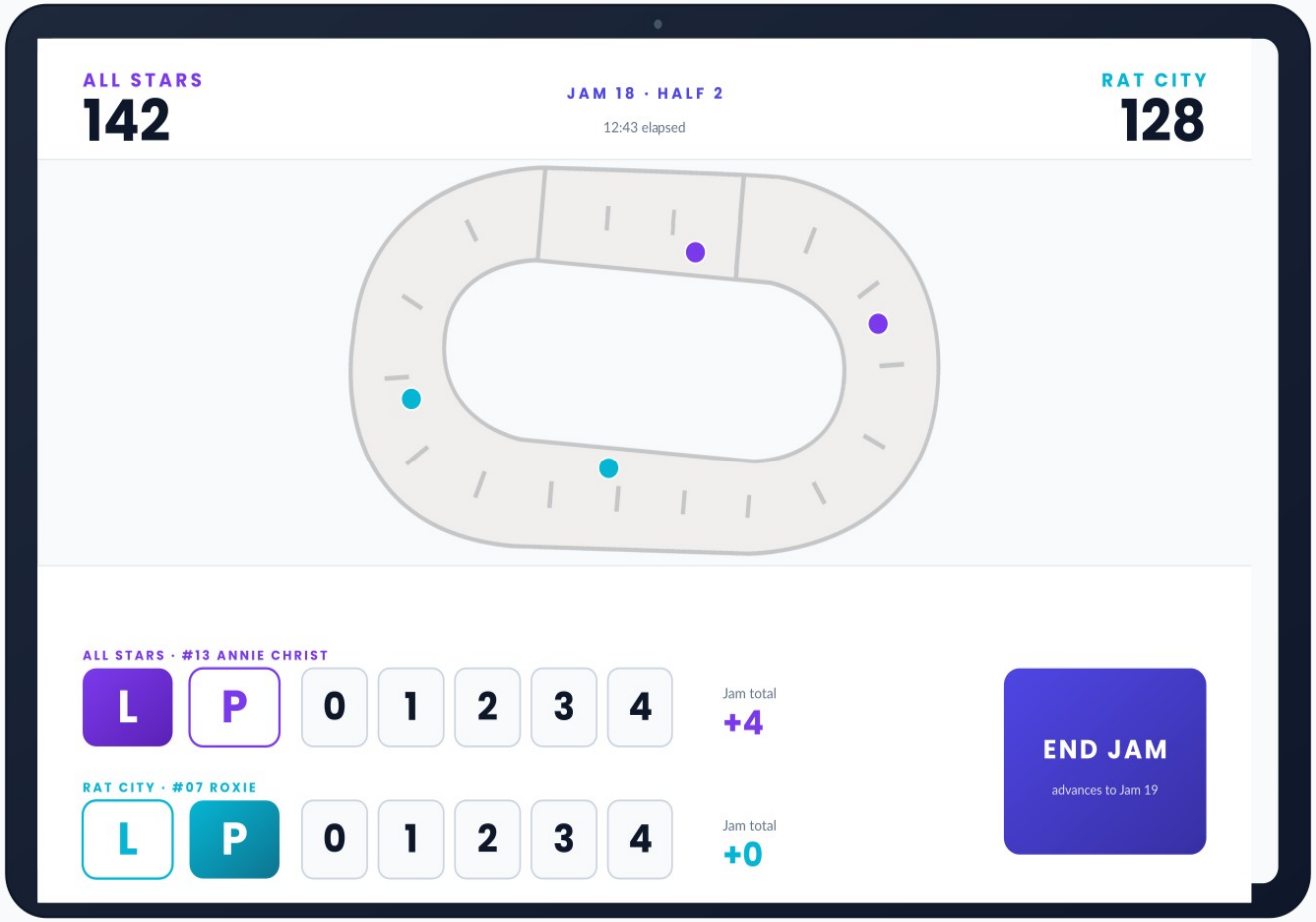
END JAM

6

One button. Loop repeats.

FEATURE 01

Live Track Tap Tracker



BOTH TEAMS, ONE SCREEN

Parallel L/P/points controls for each team. One coach tracks the whole bout.

TRACK FILLS THE TOP

Big tap targets. Built for fast, gloved-up hands.

TEAM-COLORED DOTS

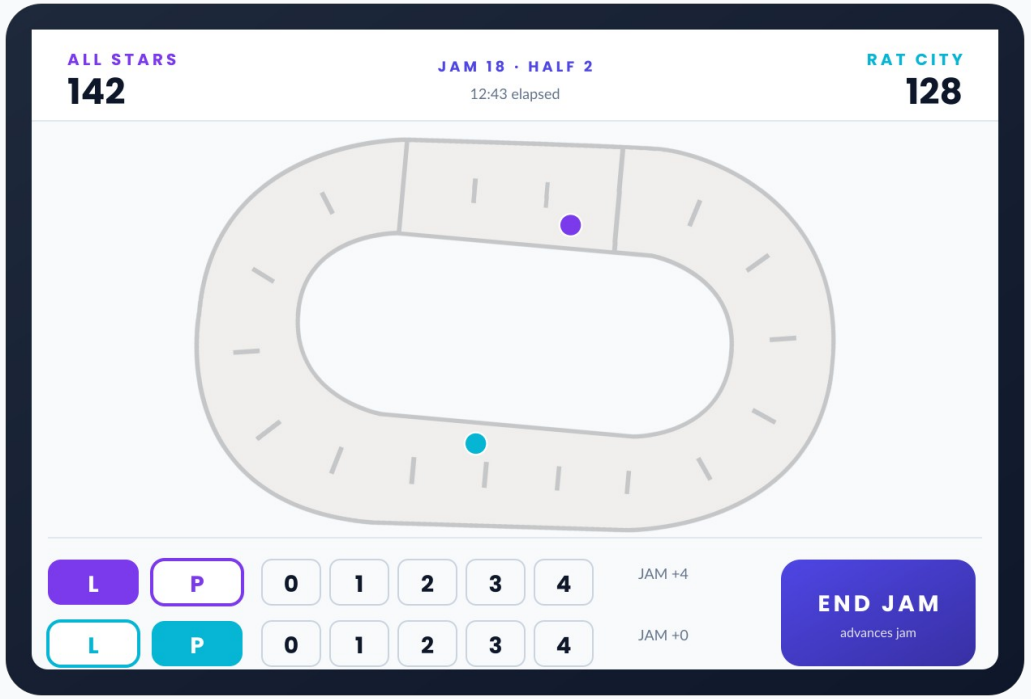
Violet = team 1. Cyan = team 2. Last 2 jams stay visible.

END JAM, ANY TIME

Single shared button. Closes the jam, advances the count.

FEATURE 02

One coach or two. Tablet or phone.



MODE 1 — SOLO

One coach. Both teams.

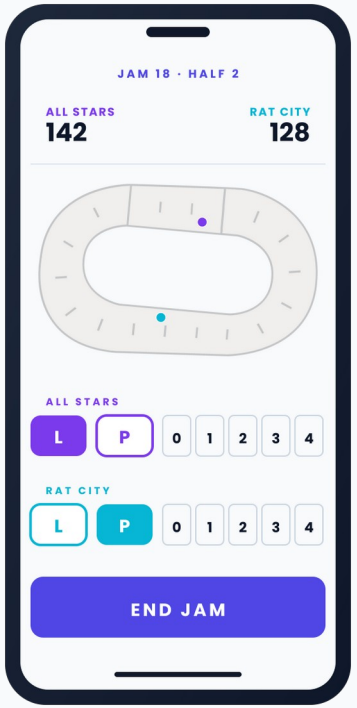
Default mode. Parallel L / P / points rows for each team — built so one coach can run the whole bout.



MODE 2 — TAG-TEAM

Two coaches. Synced.

Both tablets pick the live game from the saved list, then choose a team. You only edit yours; you see both. Local-first — merges on reconnect.



ALSO — MOBILE

Same app, on phones.

Responsive layout — when the bench has no tablet, any phone works. Track and controls stack vertically.

FEATURE 03

Lead. Star Pass. Points.

L

LEAD JAMMER

Tap when a jammer earns lead status. Drives lead percentage stats per skater across every bout.

LEAD % 67%

P

STAR PASS

Tap when the panty changes hands. Jammer auto-moves to the P slot. The new pivot scores under their original number.

STAR PASS RATE 11%

0–4

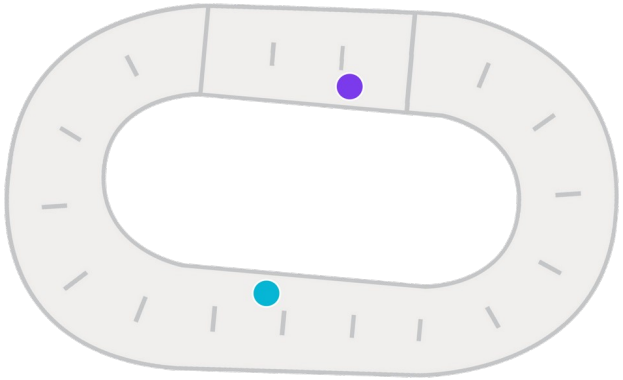
POINTS PER JAM

Tap a number for each scoring pass. We sum up the cumulative points for the jam — no per-pass admin to keep up with.

AVG / JAM 3.2

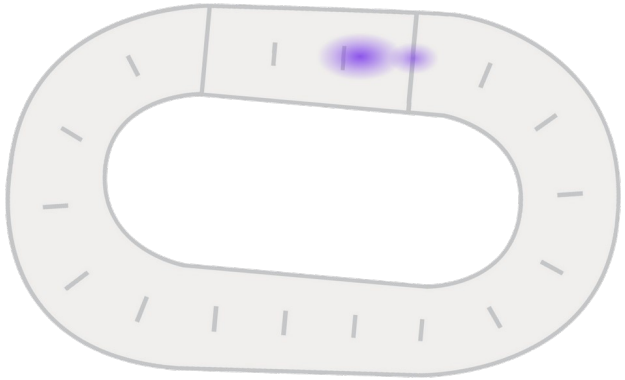
FEATURE 04

Jam Views & Heat Maps



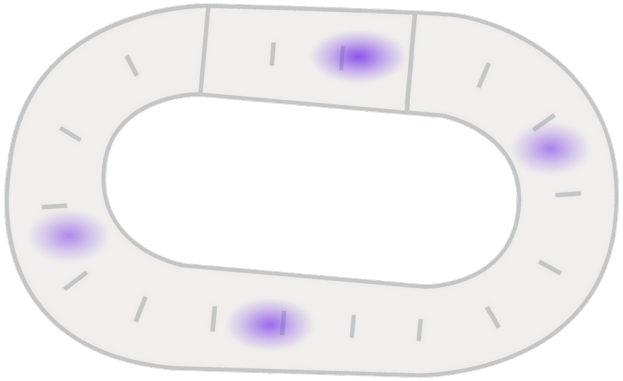
SINGLE JAM

Exact dot locations. The view you see right after a jam ends.



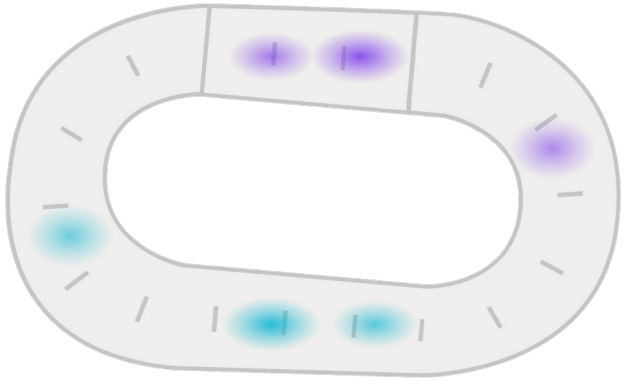
SINGLE JAMMER

One gradient. Where this skater consistently breaks the pack.



ONE TEAM

One gradient color. The team's pack-exit signature.

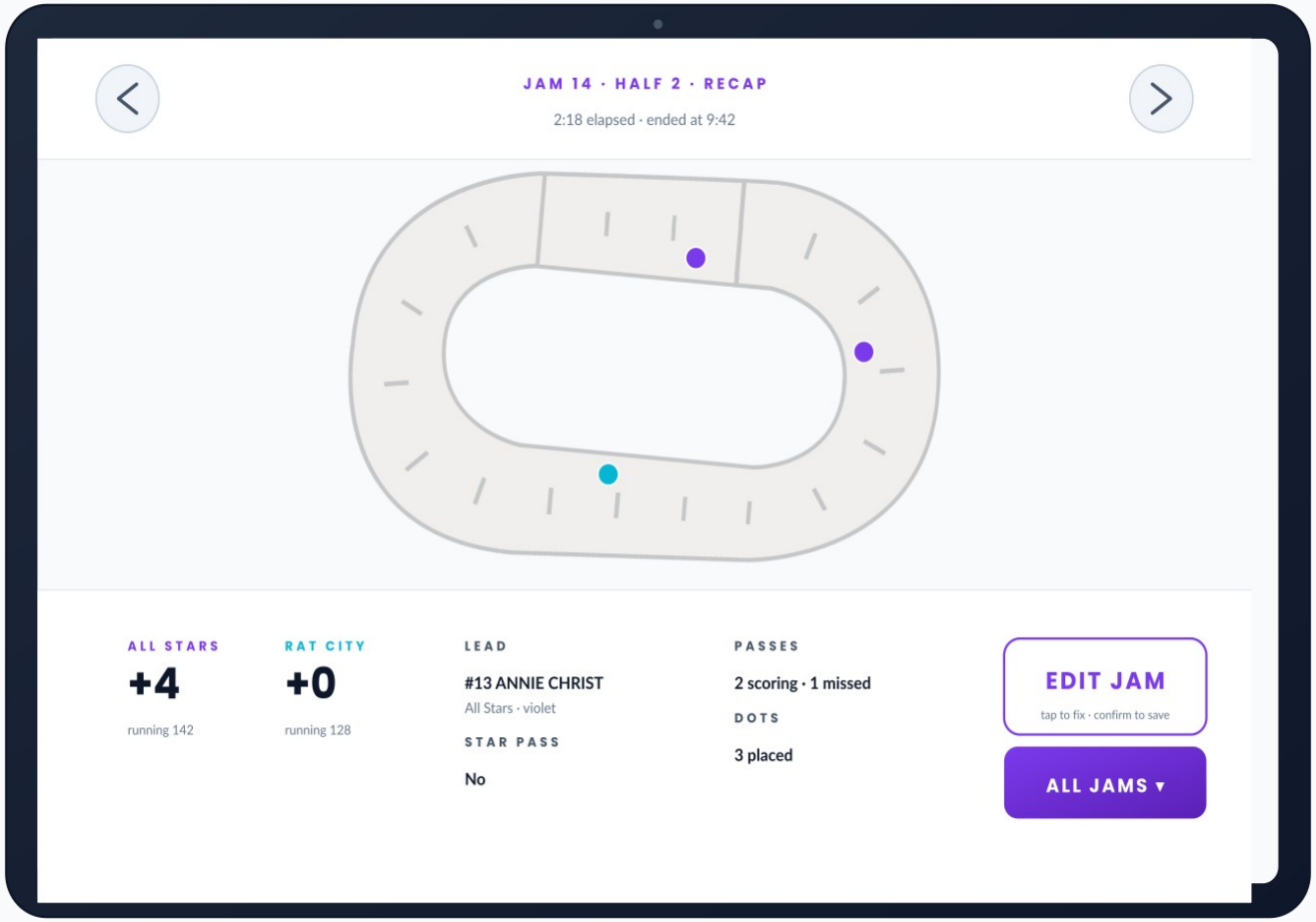


BOTH TEAMS

Two gradients overlaid. Compare patterns side by side.

FEATURE 05

Past Jams & History



SWIPE TO REWIND

Drag left or right anywhere on the track to scrub through jams. The current jam stays one tap away.

FULL HISTORY LIST

ALL JAMS opens a sortable list — every jam, both teams, points, lead, star pass, and time.

JAM RECAP VIEW

Replay the track with that jam's dots, plus the running score, lead jammer, and star-pass status.

EDIT WITH CONFIRM

Missed a tap? Open the jam, fix it, and confirm. Edits are journaled, never silently overwritten.

FEATURE 06

New Game Setup

TEAM 1

Your league's teams. Default rosters pre-loaded.

All Stars ▼

All Stars	*WFTDA Ranked*	**#12**
Grime	*WFTDA Ranked*	**#47**
Riveters	*Internal*	—
Juniors	*Internal*	—
Custom	*Type your own*	—

TEAM 2

Type-to-filter from full WFTDA list. Custom always first.

rat

Custom	*Type your own*	—
Rat City Rollergirls	*Seattle, WA*	**#28**
Rocky Mountain Rollergirls	*Denver, CO*	**#51**
Rideau Valley Vixens	*Ottawa, ON*	**#34**
Crash-Test Brats	*Newcastle, UK*	**#94**

FEATURE 07

Jammer Rosters · Both Teams

ALL STARS | RAT CITY

13	Annie Christ ✎
07	Hex Marks the Spot ✎
404	Slay Bell ✎
00	Witch, Please ✎
P	Pivot ✎

+ ADD JAMMER

Tap either tab to view, edit, add, or remove jammers for that team.

DEFAULT ROSTERS

Pre-loaded for All Stars, Grime, Riveters, Juniors. Pick a team, the jammers are already there.

OPPONENT EDITABLE

The opposing team's roster is also fully editable — add real jammer numbers as they walk out at the venue.

NUMBER FIRST

"13 — Annie Christ". Number is the primary visual; name is secondary.

NUMBERS = STRINGS

"00", "007", "3.14" — all valid. Stored as text to preserve leading zeros.

P SLOT AUTO-FILLS

Tap P during a jam → that jammer reassigns to the Pivot slot for the rest of the jam.

FEATURE 08

Score, Jams & Game History

ALL STARS	RAT CITY
142	128

RECENT GAMES *(sorted by date)*

Apr 26, 2026	All Stars vs Rat City	142–128	W
Apr 19, 2026	Grime vs Rocky Mountain	98–134	L
Apr 12, 2026	All Stars vs Rideau Valley	187–156	W
Apr 5, 2026	Riveters vs Crash-Test Brats	112–104	W

LIVE TOTALS

Both team scores always visible at the top of every screen.

JAM-BY-JAM ROW

Tap any jam to see its dot locations and edit if you mis-tapped.

DATE-SORTED HISTORY

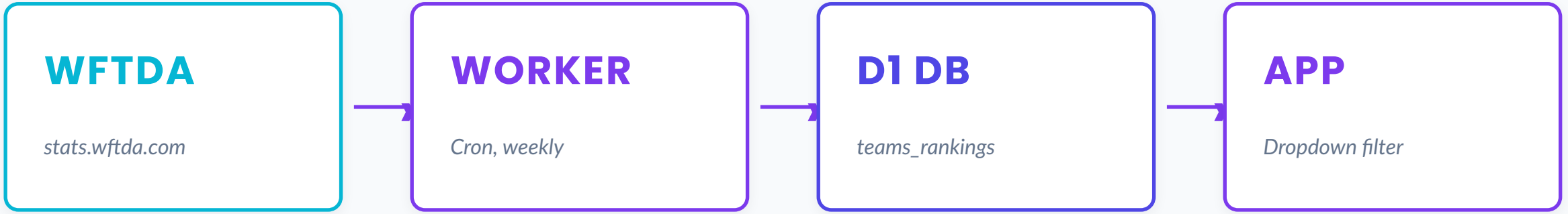
Past games stored by date + opponent. Newest first. Click into any game to re-view its data.

NO LOGIN REQUIRED

Open the link, start a game. Local-first storage syncs to D1 in the background.

FEATURE 09

WFTDA Rankings Integration



WHY SCRAPE

WFTDA doesn't expose a public ranking API. Their public page is the only source of truth.

CADENCE

Weekly cron trigger on the Worker. Rankings rarely shift mid-week — overnight Sunday is plenty.

RESILIENCE

Cache last-good snapshot in D1. If the scrape fails (markup change, downtime), serve stale data and alert.

LEGAL

Public, read-only, no scraping logged-in pages. Respect robots.txt, throttle requests, identify the User-Agent.

Six tables. SQLite on D1.

teams

```
id
name
league
wftda_rank
source
```

games

```
id
team1_id
team2_id
started_at
current_half
```

jammers

```
id
game_id
team
number (str)
name
slot
```

jams

```
id
game_id
jam_no
half
lead_team
ended_at
```

exits

```
id
jam_id
team
jammer_id
x
y
pass_no
```

scoring_passes

```
id
jam_id
team
points
by_pivot
```

Lean, edge-native, ships fast.

FRONTEND	Vite + React + TypeScript	*SPA fits live tablet use. Faster than Next.js for what we need.*
UI LIB	Material UI (MUI)	*Autocomplete carries the team picker. Solid touch-ready components.*
TRACK UI	Custom SVG + Pointer events	*Hand-built. MUI doesn't ship a derby track widget.*
HOSTING	Cloudflare Pages	*Static SPA. Free tier. Global edge. Public link, no auth.*
API	Cloudflare Workers	*Scraper, DB writes, ranking refresh.*
DATABASE	Cloudflare D1 (SQLite)	*Edge-replicated. SQL. Free for our scale.*
SOURCE	GitHub	*CI via Cloudflare's GitHub integration.*

Built for the bench, not the boardroom.

TRACK FIRST 01

The track dominates the screen. Everything else is chrome.

BIG TARGETS 02

Min 60×60pt buttons. No tiny icons. Coach is gloved-up.

ONE TAP > TWO 03

If a jam needs more than 8 taps, the design failed.

GLANCEABLE 04

Score, jam number, half — visible at all times. No drilling in.

FORGIVING 05

Mis-tap? Tap-and-hold to undo. Or fix from the jam history.

LIGHT & OFFLINE 06

Loads instantly. Works at venues with no wifi. Syncs when it's back.

What we agreed on tonight.

Workflow ✓

Coach taps L, points, P. P auto-shifts the jammer to Pivot. End-jam button closes the jam.

Auth ✓

None. Public web link, no security — no sensitive data is stored.

Game history ✓

Sorted by date, newest first. Re-open any past bout to view its jams and heat maps.

Coach mode ✓

Single coach tracks both teams from one device.

Default rosters ✓

Pre-loaded for All Stars, Grime, Riveters, Juniors. Your roster AND the opponent's are fully editable per bout.

Jammer label ✓

Number first. "13 — Annie Christ" not "Annie Christ 13".

Two nights to v1.

NIGHT 1
Tomorrow · Apr 30

Foundation + setup screens

1. **Scaffold**
   Vite + React + TS. GitHub repo. Cloudflare Pages preview deploys.
2. **Theme**
   MUI install. Build the violet/cyan/indigo theme tokens. Inter + Montserrat fonts.
3. **Track Component**
   SVG track with WFTDA markings. Pointer events drop dots in jam-coords.
4. **Game Setup**
   Two team dropdowns. League team list + Custom. WFTDA list stub for Team 2.
5. **Roster**
   Default jammers per league team. + button, edit modal, P slot logic.

NIGHT 2
Wed · May 6

Live tracking + persistence

1. **Live Tracker**
   Both-team L/P/points controls. End-jam flow. IndexedDB local-first.
2. **Heat Maps**
   heatmap.js wired. 4 view modes: single jam / single jammer / one team / both teams.
3. **D1 + Worker**
   Schema migration. /games, /jams, /exits endpoints. Sync queue.
4. **Game History**
   Date-sorted list, search by opponent, re-open past games.
5. **Polish & Ship**
   Bug pass, deploy to production Cloudflare Pages, share the link.